SIMATS ENGINEERING

ASSESSMENT TOOL 1:

Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Total Marks: 40

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Code of Conduct

I L. Sathyam Sharma (192526028 2) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L.Sathyam

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Set 2 - Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

AT-1

L. Sathyam Sharma
192526028 2
CSA01519
CLOUD COMPUTING

EXPERIMENT-1 HOSPITAL INFORMATION SYSTEM USING CLOUD

PROCEDURE

- open google Sheets and create a Spreadsheet named "SIMATS HOSPITAL"
- create colourmns: Patient name, Age, Address, Phone, Email, Attender Name, Phone, Doctor Name and Disease
- Enter sample Patients records
- create a new app to display patient record and provide forms to add, Edit and search result
- Test the application by entering a sample patient record
- Deploy or share the application with appropriate access permissions

EXPERIMENT 2 - ONLINE SUPERMARKET USING CL

PROCEDURE:

- create a google spreadsheet named online supermarket
- create a customer sheets with item name, category, price and quantity
- include categories such as Fruits, vegetables, Beverages and groceries
- open google Appsheet and create an app using the spread sheets
- Add sample problem and test the app by entering a customer order
- share or depoy the application with suitable access permission...

CREATING A VIRTUAL MACHINES USING A TYPE -2 HYPERVISOR

PROCEDURE

Daundload the Oracle Virtual Box

Download a Linux or windows OS

open and click New

Enter the virtual machine name and

Select the operating system type and Version

create the harddisk . and Finish the VM configuration

Attach the operating system ISO and optical drive

start the VM and follow the OPS installation instruction

verify the cpu and memory configura in the VM setting ..

CREATING AND RESTORING A VM SNAPSHOT AND CLONE

PROCEDURE :-

Open Virtual box and start the Virtual machine

Open the snapshot section for selected VM click Take or Take snapshot, enter the name Before changes, and save it

Start the VM and make a harmless change, such as creating a test text file

Restore the before changes snapshot and verify the test file created after the snapshot is absent

Right click the VM. and select clone

compare the cloning process and start the cloned VM to verify that it works independently

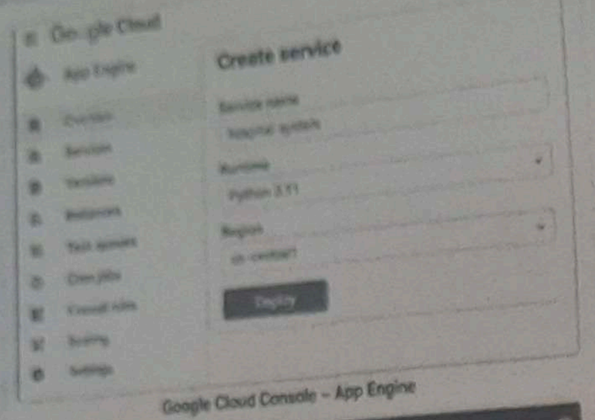
Experiment 1: Hospital Information System using Cloud (SaaS)

Procedure:

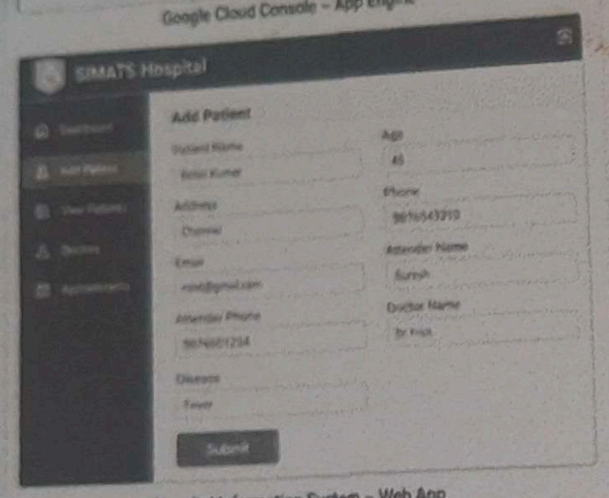
1. Create a free account on a cloud platform (e.g., Google Cloud, AWS or Azure).
2. Go to Google Cloud → Enable App Engine (or use Firebase / Cloud Run).
3. Create a simple web application using Python (Flask) or Node.js with HTML forms to store patient details.
4. Deploy the application on App Engine/Cloud Run.
5. Add patient, doctor and attendant details in the form.
6. Access the web app from the provided URL.

Tech Used:

- Google Cloud Platform (App Engine)
- Python (Flask)
- HTML, CSS, JavaScript
- Cloud Firestore / Cloud SQL (database)



Google Cloud Console – App Engine



Hospital Information System – Web App

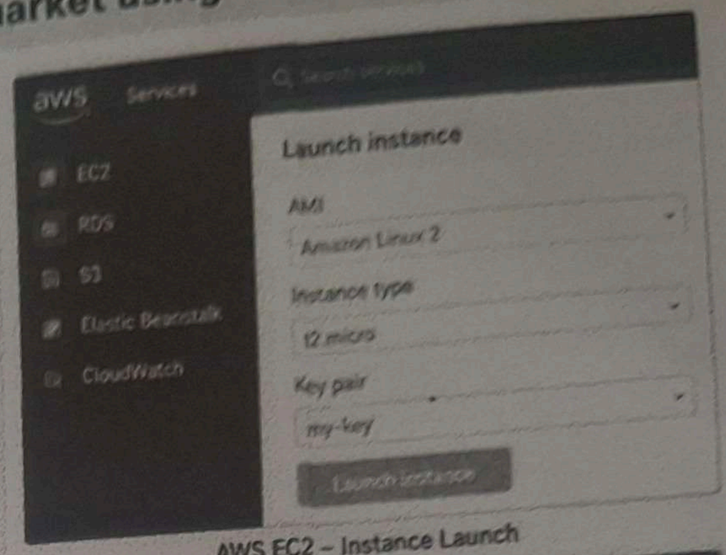
Experiment 2: Online Supermarket using Cloud (SaaS)

Procedure:

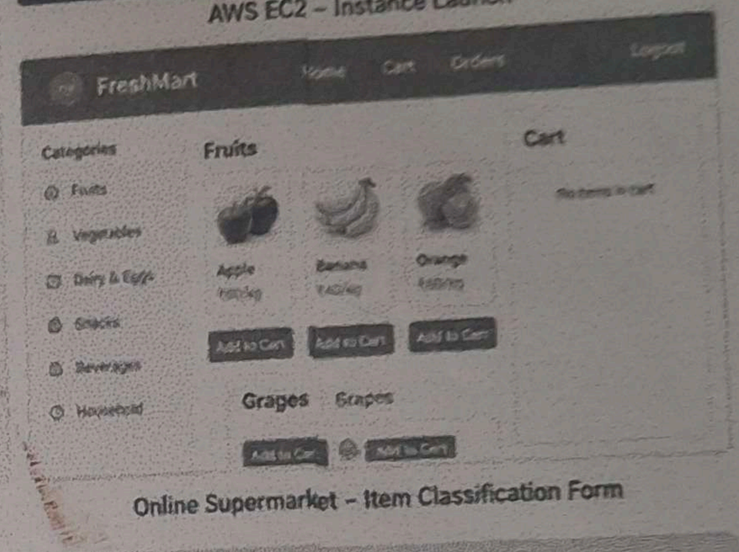
1. Create a free account on AWS (or Google Cloud).
2. Launch an EC2 instance (or use App Runner / Elastic Beanstalk).
3. Install a web server (Apache/Nginx) and deploy a simple PHP/Node.js/Flask application.
4. Design a database to store customer, item and order details.
5. Create an item classification form (e.g., Fruits, Vegetables, Beverages, etc.).
6. Access the application using the public IP/DNS.

Tech Used:

- AWS EC2 (or App Runner)
- PHP / MySQL (or Node.js + MongoDB)
- HTML, CSS, JavaScript
- AWS RDS (optional)



AWS EC2 – Instance Launch



Online Supermarket – Item Classification Form

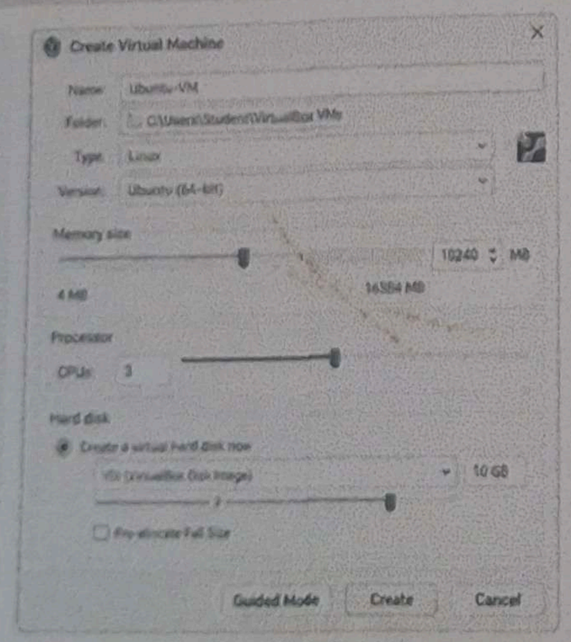
Experiment 3: Virtualization – Type 2 Hypervisor

Procedure:

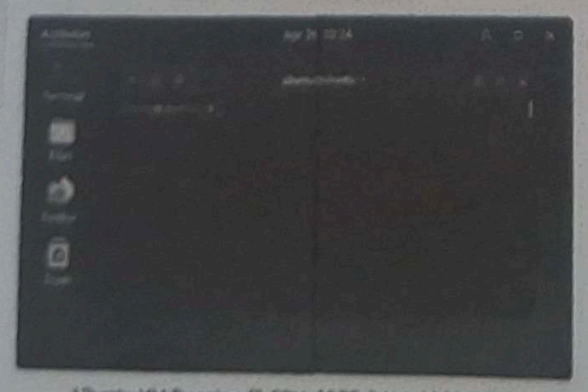
1. Install a Type 2 hypervisor (e.g., VMware Workstation Player or Oracle VM VirtualBox).
2. Create a new virtual machine (VM).
3. Configure the VM with:
   - 3 CPU cores
   - 10 GB RAM
   - 10 GB storage disk
4. Install the host OS (e.g., Ubuntu / Windows).
5. Start the VM and verify the configuration.

Tech Used:

- VirtualBox / VMware Workstation Player
- Ubuntu 22.04 (or Windows 10/11)
- VM settings (3 CPU, 10GB RAM, 10GB Disk)



VirtualBox – VM Configuration



Ubuntu VM Running (3 CPU, 10GB RAM, 10GB Disk)

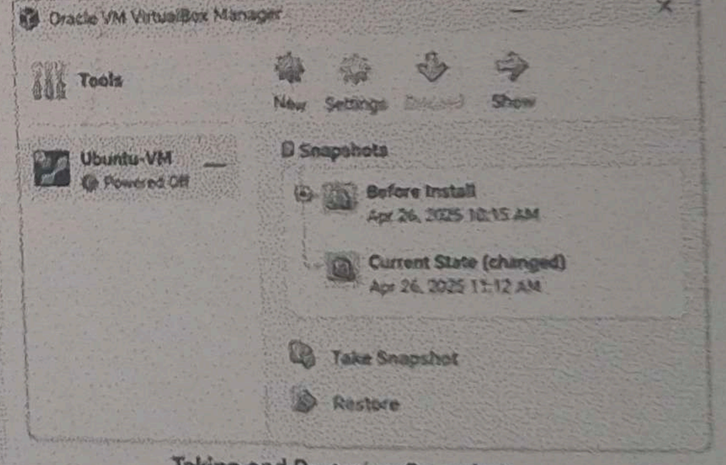
Experiment 4: Virtualization – Snapshot and Clone

Procedure:

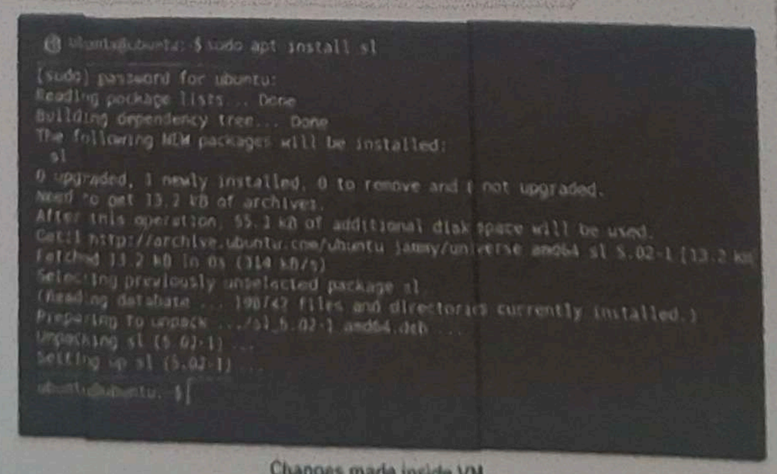
1. Start the existing VM (from Exp. 3).
2. Take a snapshot of the VM (e.g., before installing an application).
3. Make changes inside the VM (e.g., install a package or create a file).
4. Revert to the snapshot and check that the changes are removed.
5. Create a clone of the VM.
6. Start the cloned VM and verify it works independently.

Tech Used:

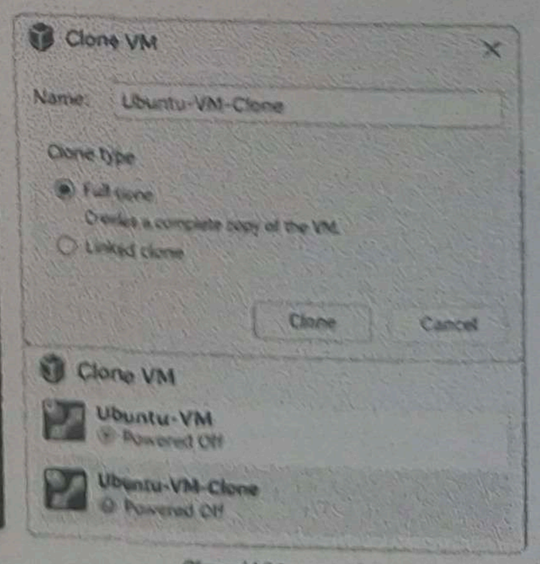
- VirtualBox / VMware Workstation Player
- Snapshot & Clone features



Taking and Restoring Snapshot



Changes made inside VM



Cloned VM created